

ABOUT ME

I am a Ph.D. candidate at the University of Michigan studying both computer and cognitive science. I study large language models from a cognitive perspective, examining the extent to which their processes of knowledge acquisition resemble those of humans.

EDUCATION

The University of Michigan Ph.D. in Computer Science and Engineering, Certificate in Cognitive Science	Ann Arbor, MI 2024–Current
The University of Michigan M.S. in Computer Science and Engineering	Ann Arbor, MI 2022–2024
The University of Texas B.S. in Computational Engineering, Certificate in Evidence and Inquiry	Austin, TX 2016–2020

FELLOWSHIPS AND AWARDS

• Outstanding Paper Award at AIA Workshop @ COLM	Fall 2025
• National Science Foundation Graduate Research Fellowship	Winter 2024
• CSE Department Outstanding Graduate Student Instructor Award	Fall 2023
• CSE Department Outstanding Graduate Student Instructor Award	Winter 2023
• Northrop Grumman BRAVO to our Stars (3x)	2021–2022
• FSTI Award for Excellence in Chemistry	Spring 2018
• TIDES Advanced Summer Research Fellowship	Summer 2017
• Engineering Honors Scholarship	2016–2020
• Polymathic Scholars Interdisciplinary Humanities and Natural Science Honors	2016–2020

CONFERENCE PUBLICATIONS

[C1] Y. Fu, R. Qiu, X. Wang, *et al.*, “Beyond blind following: Evaluating robustness of llm agents under imperfect guidance”, in *EACL*, 2026.

[C2] **J. Sansom**, M. Khalifa, S. Wu, H. Lee, and J. Chai, “Show or tell? interactive task learning with large language models”, in *Submission*, 2026.

[C3] A. Z. Liu, X. Wang, **J. Sansom**, *et al.*, “Interactive and expressive code-augmented planning with large language models”, in *ACL*, 2025.

[C4] Y. Huang, **J. Sansom**, Z. Ma, F. Gervits, and J. Chai, “DriVLMe: Exploring Foundation Models as Autonomous Driving Agents That Perceive, Communicate, and Navigate”, in *IROS*, 2024.

[C5] Z. Ma, **J. Sansom**, R. Peng, and J. Chai, “Towards A Holistic Landscape of Situated Theory of Mind in Large Language Models”, in *Findings of EMNLP*, 2023.

WORKSHOP PUBLICATIONS

- [W1] Y. Fu, R. Qiu, X. Wang, *et al.*, “Beyond blind following: Evaluating robustness of llm agents under imperfect guidance”, in *AIA Workshop @ COLM*, 2025.
- [W2] **J. Sansom**, M. Khalifa, H. Lee, and J. Chai, “Show or tell? interactive task learning with large language models”, in *Multi-Turn Interactions in LLMs Workshop @ NeurIPS*, 2025.

JOURNAL PUBLICATIONS

- [J1] E. Lejeune, A. Khang, **J. Sansom**, and M. Sacks, “FM-Track: A Fiducial Marker Tracking Software for Studying Cell Mechanics in a Three-Dimensional Environment”, in *SoftwareX 11*, 2020, p. 100417.
- [J2] A. Khang, A. Rodriguez, M. Schroeder, **J. Sansom**, E. Lejeune, and M. Sacks, “Quantifying Heart Valve Interstitial Cell Contractile State Using Highly Tunable Poly(Ethylene Glycol) Hydrogels”, in *Acta Biomaterialia 96*, 2019, pp. 354–367.

INDUSTRY EXPERIENCE

LG AI Research

Ann Arbor, MI

Research Intern

2023–2024

- Crowdsourced more than 10,000 examples of people using the internet for AI model training
 - * Created a Chromium extension for recording browser interactions and a server for hosting virtual machines
 - * Automated task creation and quality checks of crowdsourced data

Northrop Grumman

San Diego, CA

Systems Engineer (Technical Level II), Pathways Rotational Training Program

2020–2022

- Leveraged my expertise in the HW-, SW-, and algorithm-level architecture of a fielded, software-defined radio to:
 - * Assist a cross-organizational team with the design and deployment of a novel DSP algorithm
 - * Author and obtain customer funding for a proposal detailing improvements to a fielded DSP algorithm
- Created the AI Corporate Catalog, a company-wide database of AI/ML capabilities
- Led a small team in the design and deployment of a C++ unit testing infrastructure

Ansys Government Initiatives

Exton, PA

Corporate Systems Engineering Intern

Summer 2019

- Used Python to quantify the accuracy of orbital decay forecasts in STK, AGI’s primary software offering
- Helped develop multiple simulations that modeled orbital dynamics, communications links, and terrain effects
- Outlined a strategy to bolster STK’s collaborative capabilities and presented it to the senior development team

TEACHING EXPERIENCE

- **Graduate Student Instructor** at the University of Michigan (**Outstanding GSI Award**) Winter 2023
Introduction to Natural Language Processing (EECS 487)
- **Graduate Student Instructor** at the University of Michigan (**Outstanding GSI Award**) Fall 2023
Introduction to Natural Language Processing (EECS 487)

PRESENTATIONS

- **J. Sansom** “All-or-Nothing Thinking: Large Language Models and Generic Statements,” presented at the University of Michigan’s Cognitive Science Seminar. October 6, 2026; Ann Arbor, MI.
- L. Girouard and **J. Sansom** “Seminar on AI and Education,” presented at an elementary school PTA meeting. April 27th, 2026; Livonia, MI.
- **J. Sansom** “Investigating Methodology for Global Optimization,” presented at the College of Natural Sciences Undergraduate Research Forum. April 13th, 2018; Austin, TX. **(FSTI Award for Excellence in Chemistry)**

REVIEWING EXPERIENCE

- **Conferences:** COLM '26, ICML '25
- **Workshops:** AI4Good @ ICML '26, SCALR @ COLM '25, Multi-Turn Interactions in LLMs @ NeurIPS '25

SKILLS

- **Languages:** Python, C++, JavaScript, HTML, MATLAB, Bash
- **Software Tools:** PyTorch, Jax, NLTK, Transformers, W&B, Scikit-Learn, NumPy, SciPy, Git, Docker, OpenMP, Selenium, Playwright, Flask

EXTRACURRICULAR ACTIVITIES

- Graduate Student Recruitment Chair at the University of Michigan CSE Department 2026–2027
Will lead graduate student participation in the recruitment of new students
- Chair of Northrop Grumman Pathways Professional Development Committee 2021–2022
Planned and successfully launched a new technical mentorship program for early-career engineers
- Volunteer at the Arc and the Rosedale School 2018–2019
Helped adults and children with cognitive disabilities develop life skills and provided constant positive feedback
- Eagle Scout and Troop Guide in the Boy Scouts of America 2016
Led a team of 30 to construct shelves for a homeless shelter. Taught younger scouts various scouting skills